

Cover Letter — PLOS ONE

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Title: When a model is not retained: a pre-registered out-of-sample audit standard for structural modules in empirical environmental forecasting **Author:** Amin Abaee (Independent Researcher) **Journal:** *PLOS ONE* **Manuscript carrier:** paperF1_retention_framework_v30_humanized (typst stamp se30-c1375ed), with Supplement S1 v30 **Keywords:** forecast evaluation; model retention; pre-registration; negative results; environmental forecasting; stock assessment; groundwater; operating characteristics

Dear Editor,

I am pleased to submit my manuscript, “**When a model is not retained: a pre-registered out-of-sample audit standard for structural modules in empirical environmental forecasting**,” for consideration in *PLOS ONE*.

What the paper does. Environmental forecasting practice—across fisheries, water resources, and beyond—builds structural modules by default, yet there is no pre-registered standard that says when added structure has earned its place out-of-sample. This paper supplies and demonstrates one: a frozen retention rule in which a candidate causal module is retained only if it reduces rolling-origin RMSE against both last-value persistence and the next-simpler rung by more than a stated 5% tie band, at both one-step and multi-step horizons, with information availability documented per input at each forecast origin. The rule is applied to two unrelated, management-demanding empirical cases—Northern cod biomass (NAFO 2J3KL, two assessment series) and Edwards Aquifer head at index well J-17 (1934–2023)—with per-origin forecast files archived so every verdict can be recomputed.

Why it matters. The two worked examples reach empty retained sets for opposite reasons: on cod, the closest structural approach misses the band at +9.01% (five years) and +17.09% (one year), never approaching retention, while on Edwards the best-scoring module passes the predictive gates but is declined on pre-specified class grounds (it collapses to an affine AR(1) map). The paper then measures the instrument itself against known ground truth, because a negative verdict is informative only if the gate has teeth where truth is strong. On 10,000 pre-registered simulation replicates the rule is highly specific (0.978 under a persistence null), powerful only where signal is strong (0.955–0.960 at collapse-window parameters), and weak against stock-flow and depensatory dynamics—power is low for three of four in-class processes, and the paper shows the cause is parameter identification (likelihood flatness and compensation), not gate conservatism: removing all gates adds almost no power. Out-of-class truths (time-varying productivity, observation error) produce false retention at 0.633–0.933, delimiting what non-retention can and cannot certify. A comparison against rule variants on the same archived replicates (information criterion: 0.509/0.992/0.615; scaled-error gate: 0.651/0.675/0.945; the rule as stated: 0.376/0.978/0.835) locates the trade the pre-registered rule makes, and the verdicts are shown band-invariant, with a simulation-calibrated band (power ≥ 0.80 , specificity ≥ 0.90) registered exclusively for future applications—it never re-opens the verdicts reported here.

Why it is appropriate for *PLOS ONE*. *PLOS ONE* evaluates rigour and methodology rather than perceived impact, and explicitly welcomes well-executed negative results—exactly what this paper is: a methodological standard proven out on two hard real datasets, calibrated against known synthetic truth, every intermediate classification reproducible from registered files. The contribution is a certified instrument for auditing structural complexity claims in any forecast-driven environmental discipline, together with its measured limits: which failure modes it detects, which it cannot, and where a stronger band is licensed only prospectively.

That combination—pre-registration, conflict-free negative certificates scoped to estimator and series, byte-level reproducibility, and honest power accounting—sits at the centre of the journal’s reproducibility and methodology remit.

Fit and originality. To my knowledge this is the first pre-registered forecast-retention audit standard applied jointly across fisheries and groundwater with archived per-origin replicates, likelihood-ratio evidential weights per generating mechanism, and a calibrated prospective band registered before use. The paper’s claims are conservative throughout: it makes no sustainability statement about either system, draws no unlicensed physical inference from predictive retention, and states precisely which conclusions are licensed, which are open, and which are ruled out.

Declarations. All input data, analysis campaigns, frozen result files, the 10,000-replicate archives, and the independent band-calibration audits are archived in a public repository (<https://github.com/MIKEAA2020/general-sustainability>) and regenerate deterministically, byte for byte. The work is original, is not under consideration elsewhere, and I have no competing interests. Funding: none received.

Thank you for your consideration.

Yours sincerely,

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